# Spectral methods for homogenization

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Joint work with Éric Cancès and David Gontier

September 26$^{\text{th}}$ 2022, Aspect'22, Oldenburg

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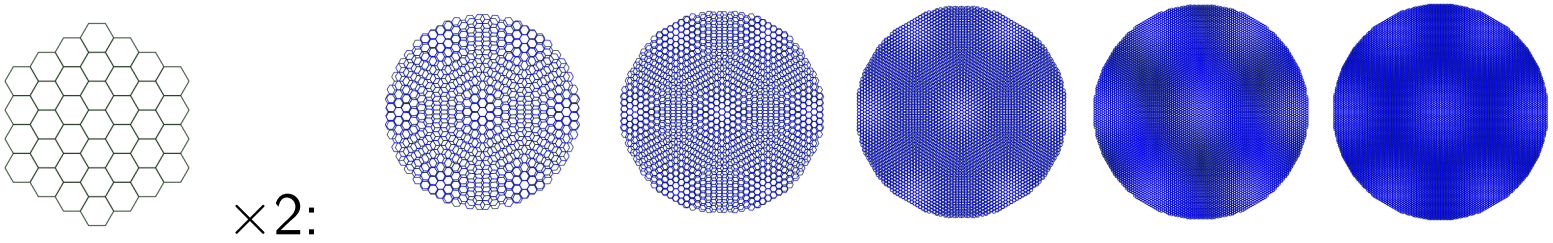

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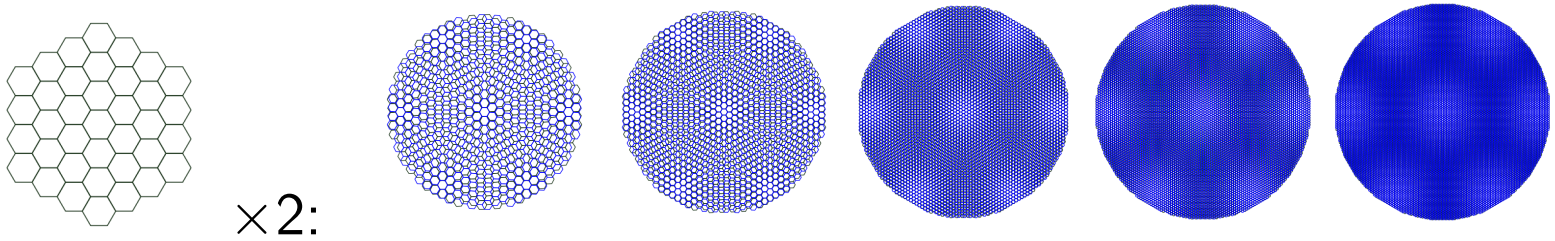


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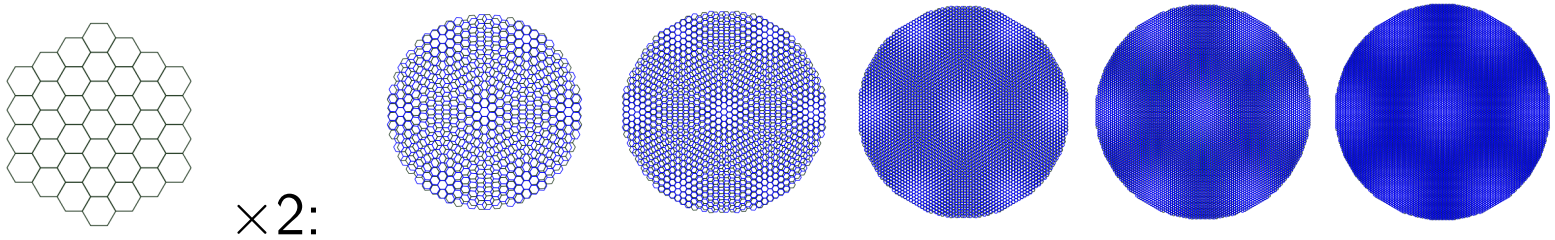


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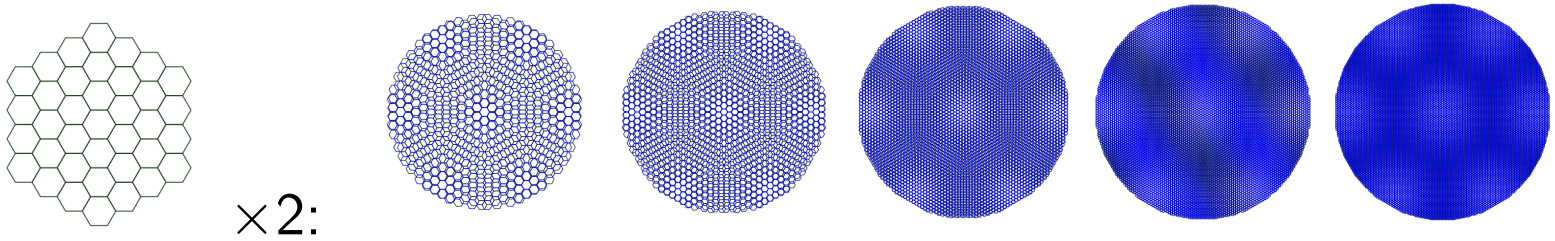


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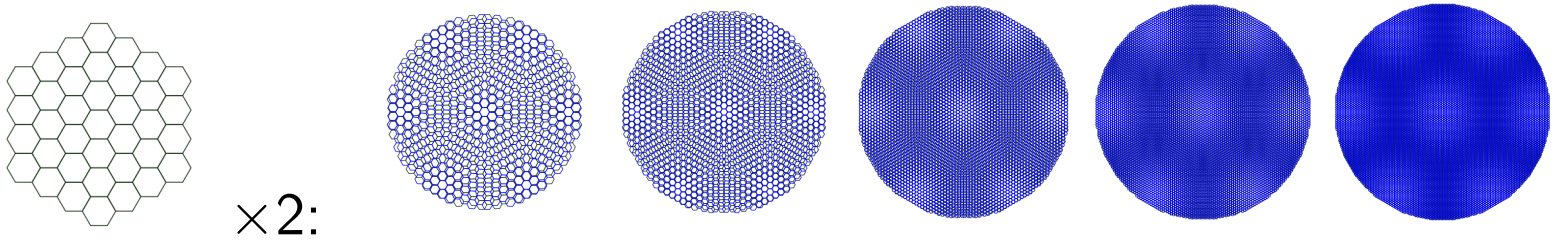


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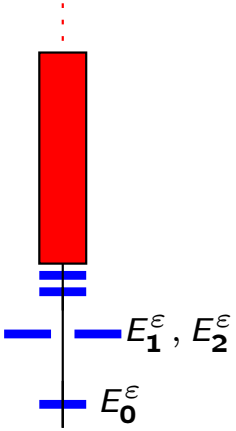
- potential of rescaled electric charge densities, modelizes high pressures
- with $\alpha \mapsto -\Delta + \frac{1}{\varepsilon^\alpha} v\left(x, \frac{x}{\varepsilon}\right)$, the critical regime is $\alpha = 1$
- similar methods can be applied to $H^\varepsilon = \operatorname{div} A\left(x, \frac{x}{\varepsilon}\right) \nabla$

# Definition of eigenmodes




$\sigma\left(H^\varepsilon\right)$

Eigenvalues

$$E_\ell^\varepsilon := \sup_{\substack{A \subset L^2_{\mathrm{per}}(\Omega) \\ \dim_{\mathbb{C}} A = \ell}} \inf_{\substack{\Psi \in A^\perp \\ \int |\Psi|^2 = 1 \\ \Psi \in H^1}} \langle \Psi, H^\varepsilon \Psi \rangle$$

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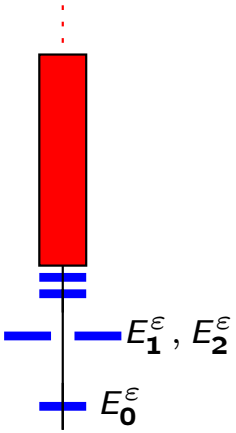


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Eigenfunctions : $H^\varepsilon \Psi_\ell^\varepsilon = E_\ell^\varepsilon \Psi_\ell^\varepsilon,\ \Psi_\ell^\varepsilon \in H^1_{\mathrm{periodic}}(\Omega, \mathbb{C})$

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# Existing results

Bensoussan-Lions-Papanicolau 78, Duchêne-Raymond 18, Zhang 21, LeBris-Goudey 22

$$\Delta_y \chi(x,y) \underset{\forall x}{=} v(x,y), \qquad V_0(x) := -\frac{1}{|\Omega|}\int_\Omega |\nabla_y \chi_1(x,y)|^2 \,\mathrm{d}y$$

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(Linear equation problem) Take $\mu < \inf \sigma\left(H^0\right)$, define $(H^\varepsilon - \mu)u^\varepsilon = f$, $(H^0 - \mu)u^0 = f$, then

$$\left\| u^\varepsilon(x) - \left(1 + \varepsilon\chi_1\left(x, \tfrac{x}{\varepsilon}\right)\right) u^0(x) \right\|_{H^1} \leqslant c\varepsilon$$

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(Eigenmodes problem in dimension $d = 1$) Define $(H^\varepsilon - E_\ell^\varepsilon)\Psi_\ell^\varepsilon = 0$, $(H^0 - E_\ell^0)\Psi_\ell^0 = 0$, then

$$\left\| \Psi_\ell^\varepsilon(x) - \left(1 + \varepsilon\chi_1\left(x, \tfrac{x}{\varepsilon}\right)\right) \Psi_\ell^0(x) \right\|_{H^1} \leqslant c\varepsilon, \qquad \left| E_\ell^\varepsilon - E_\ell^0 \right| \leqslant c\varepsilon.$$

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# Goals

Obtain second-order correctors $C_\varepsilon$

- (Linear equation $(H^\varepsilon - \mu)u^\varepsilon = f$)

$$\left\| u^\varepsilon - C_\varepsilon u^0 \right\|_{H^1} \leqslant c\varepsilon^2$$

- (Eigenmodes $H^\varepsilon \Psi_\ell^\varepsilon = E_\ell^\varepsilon \Psi_\ell^\varepsilon$)

$$\left\| \Psi_\ell^\varepsilon - C_\varepsilon \Psi_\ell^0 \right\|_{H^1} \leqslant c\varepsilon^2, \qquad \left| E_\ell^\varepsilon - \left( E_\ell^0 + \varepsilon E_\ell^1 \right) \right| \leqslant c\varepsilon^2$$

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# Main result

Recall that $H^\varepsilon := -\Delta + \frac{1}{\varepsilon} v\left(x, \frac{x}{\varepsilon}\right)$, define the correctors $\chi_1(x,y), \chi_2(x,y), A_2(x,y)$,

$$\Delta_y \chi_1 = v, \qquad \Delta_y A_2 := -2\nabla_y \chi_1,$$
$$\Delta_y \chi_2 := \chi_1 v - V - 2\nabla_x \cdot \nabla_y \chi_1,$$

the potentials $V_0(x) := -\frac{1}{|\Omega|}\int_\Omega |\nabla_y \chi_1|^2 \,\mathrm{d}y$,

$$V_1(x) := \frac{2}{|\Omega|}\int_\Omega \left(\nabla_x \chi_1 \cdot \nabla_y \chi_1 - \chi_1 |\nabla_y \chi_1|^2\right) \mathrm{d}y$$

the operator $H^0 := -\Delta + V_0(x)$ and the corrector operator

$$\mathcal{M}^\varepsilon := 1 + \varepsilon \chi_1\left(x, \tfrac{x}{\varepsilon}\right) + \varepsilon^2\left(\chi_2\left(x, \tfrac{x}{\varepsilon}\right) + A_2\left(x, \tfrac{x}{\varepsilon}\right) \cdot \nabla\right)$$

Theorem : for any $z \in \mathbb{C} \backslash \sigma\left(H^0\right)$,

$$\boxed{\left\| (H^\varepsilon - z)^{-1} - \mathcal{M}^\varepsilon \left((H^0 + \varepsilon V_1) - z\right)^{-1} \right\|_{H^1 \to H^1} \leqslant C(z)\varepsilon^2}$$

# Proof

Compute

$$(H^\varepsilon - z)\,\mathcal{M}^\varepsilon = H^0 + \varepsilon V_1 - z + \varepsilon C_\varepsilon$$

by computing commutators $[\Delta, A\left(x, \frac{x}{\varepsilon}\right) \cdot \nabla]$, ...

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Then divide by $H^\varepsilon - z$ on the left, and by $H^0 + \varepsilon V_1 - z$ on the right, and show that

$$\left\| (H^\varepsilon - z)^{-1}\, C_\varepsilon \left( (H^0 + \varepsilon V_1) - z \right)^{-1} \right\|_{H^1 \to H^1} \leqslant c\varepsilon$$

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# Linear equation

Consider $(H^\varepsilon - z)u^\varepsilon = f$ and $(H^0 + \varepsilon V_1 - z)u^{0,\varepsilon} = f$, then

$$\left\| u^\varepsilon - \mathcal{M}^\varepsilon u^{0,\varepsilon} \right\|_{H^1} \leqslant C\varepsilon^2 \left\| f \right\|_{H^1}$$

**Proof.**

$$u^\varepsilon - \mathcal{M}^\varepsilon u^{0,\varepsilon} = \left( (H^\varepsilon - z)^{-1} - \mathcal{M}^\varepsilon (H^0 + \varepsilon V_1 - z)^{-1} \right) f \qquad \square$$

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$$\|\Psi^\varepsilon - \mathcal{M}^\varepsilon \varphi^\varepsilon\|_{H^1} \leqslant C\varepsilon^2, \qquad \text{so } \left|E_\ell^\varepsilon - E_\ell^{\varepsilon,(2)}\right| \leqslant C\varepsilon^4$$

where $E_\ell^{\varepsilon,(2)} := \left\langle \frac{\mathcal{M}^\varepsilon \varphi^\varepsilon}{\|\mathcal{M}^\varepsilon \varphi^\varepsilon\|_{L^2}}, H^\varepsilon \frac{\mathcal{M}^\varepsilon \varphi^\varepsilon}{\|\mathcal{M}^\varepsilon \varphi^\varepsilon\|_{L^2}} \right\rangle$

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**Proof.**

Uses Cauchy's resolvent formula,

$$P_{\Psi^\varepsilon} - \mathcal{M}^\varepsilon P_{\varphi^\varepsilon} = \frac{1}{2\pi i} \int_{\mathcal{C}} \mathrm{d}z \left( (H^\varepsilon - z)^{-1} - \mathcal{M}^\varepsilon (H^0 + \varepsilon V_1 - z)^{-1} \right)$$

□

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# Simulations

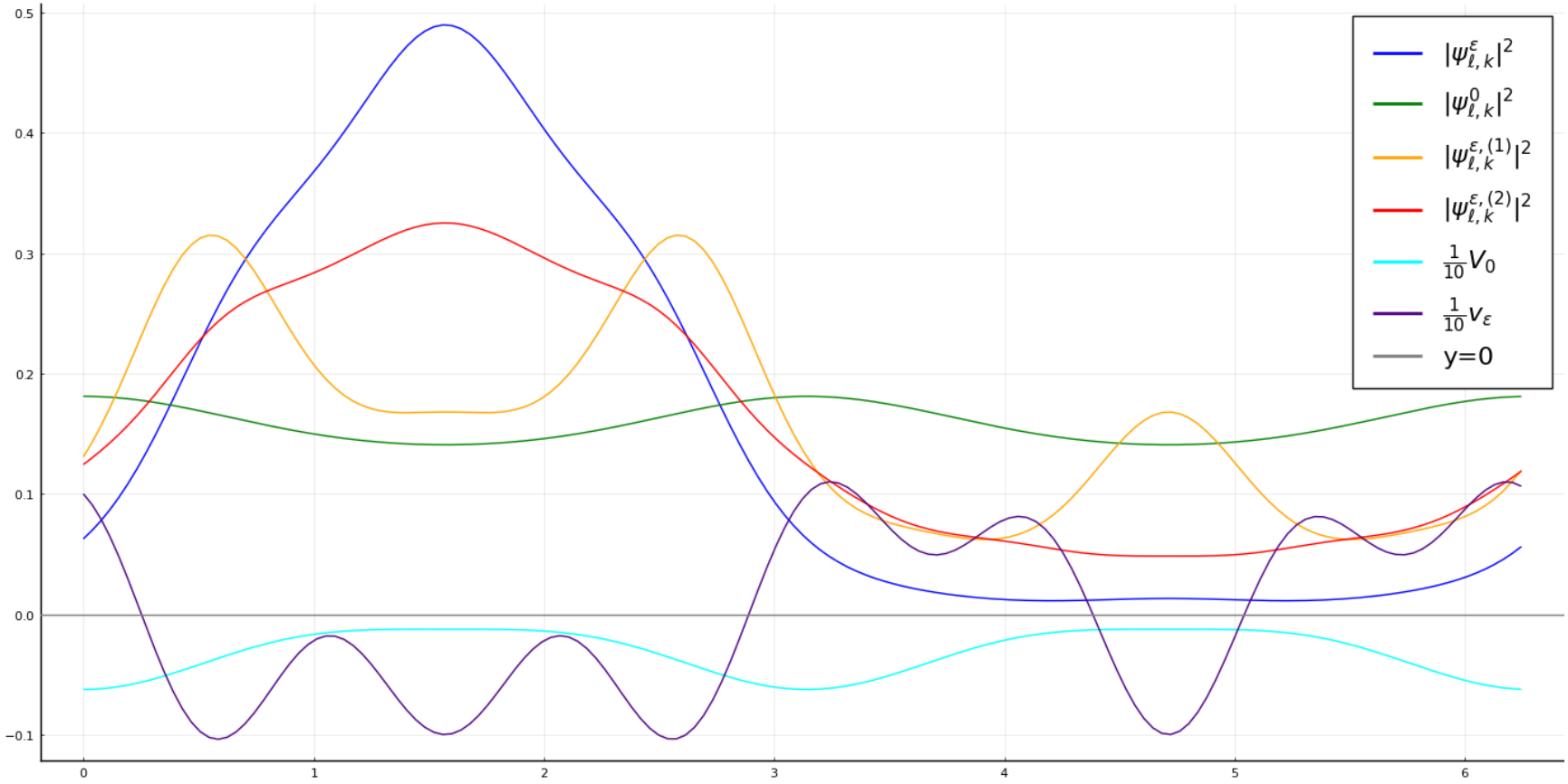


Figure: Potentials and eigenfunctions densities for $\varepsilon = \frac{1}{2}$, and $v(x,y) = \cos(2x)\cos(2y) - \cos(x)\,\sin(y)$

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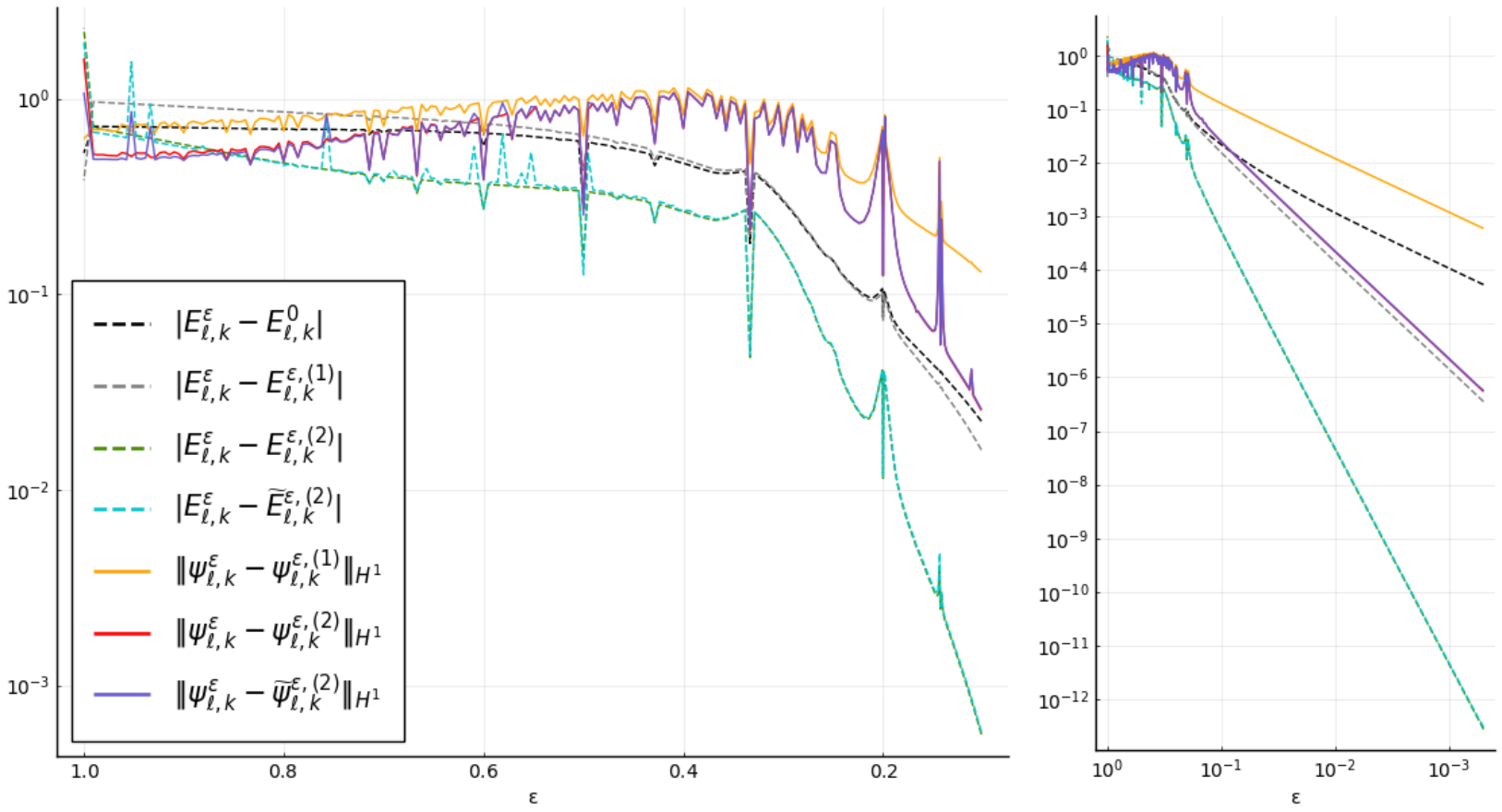


Figure: Errors for $\varepsilon \in [10^{-1}, 1] \cap \mathbb{Q}$ (left), and $\varepsilon \in [10^{-3}, 1] \cap \mathbb{Q}$ (right)

# Delocalization

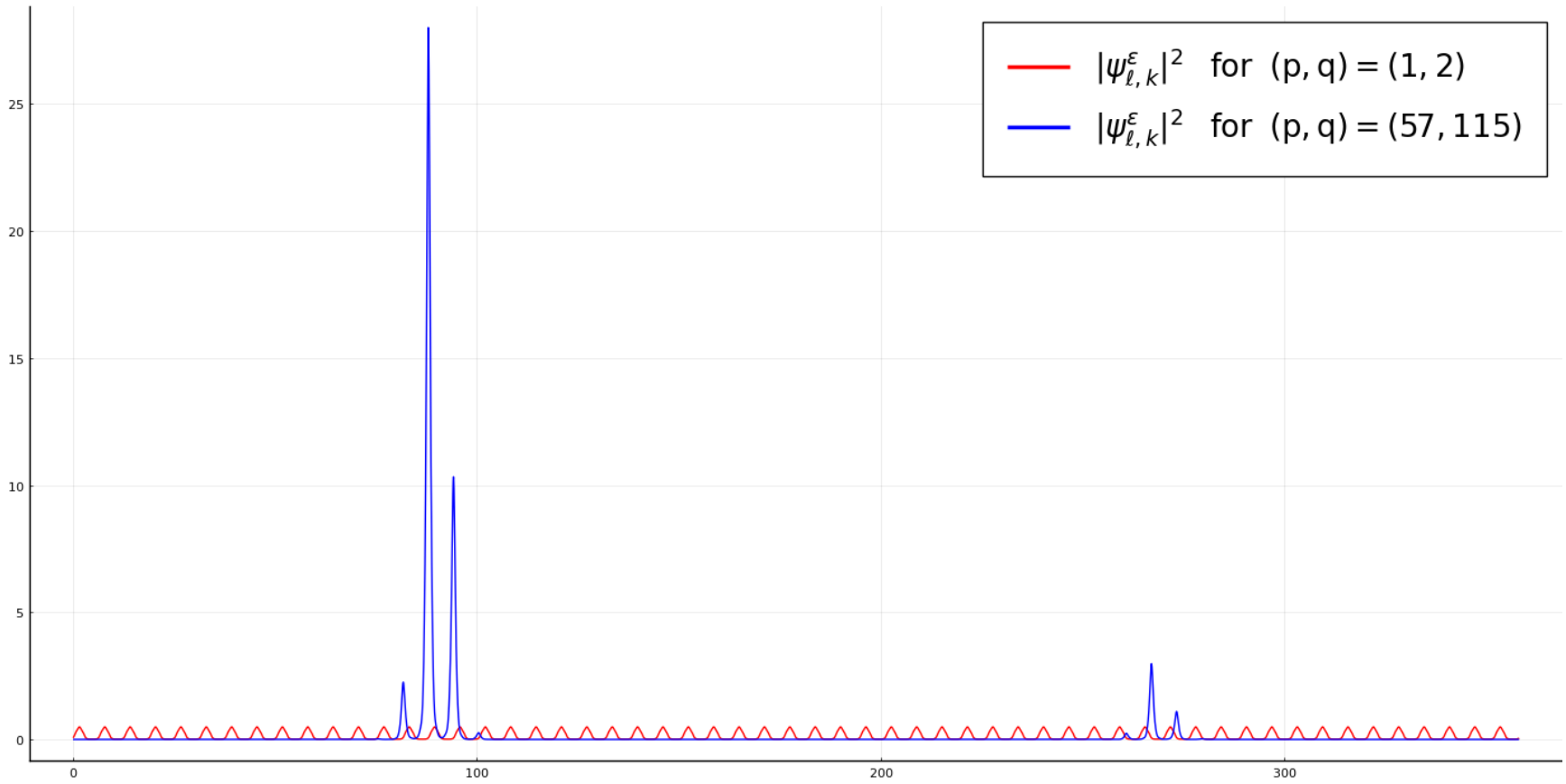


Figure: Plots of $|\Psi^{\varepsilon}|^2$ for $\varepsilon = \frac{1}{2}$ and $\varepsilon = \frac{57}{115} = \frac{1}{2} - \frac{1}{330}$ on their smallest common periodic cell $[0, 114\pi]$

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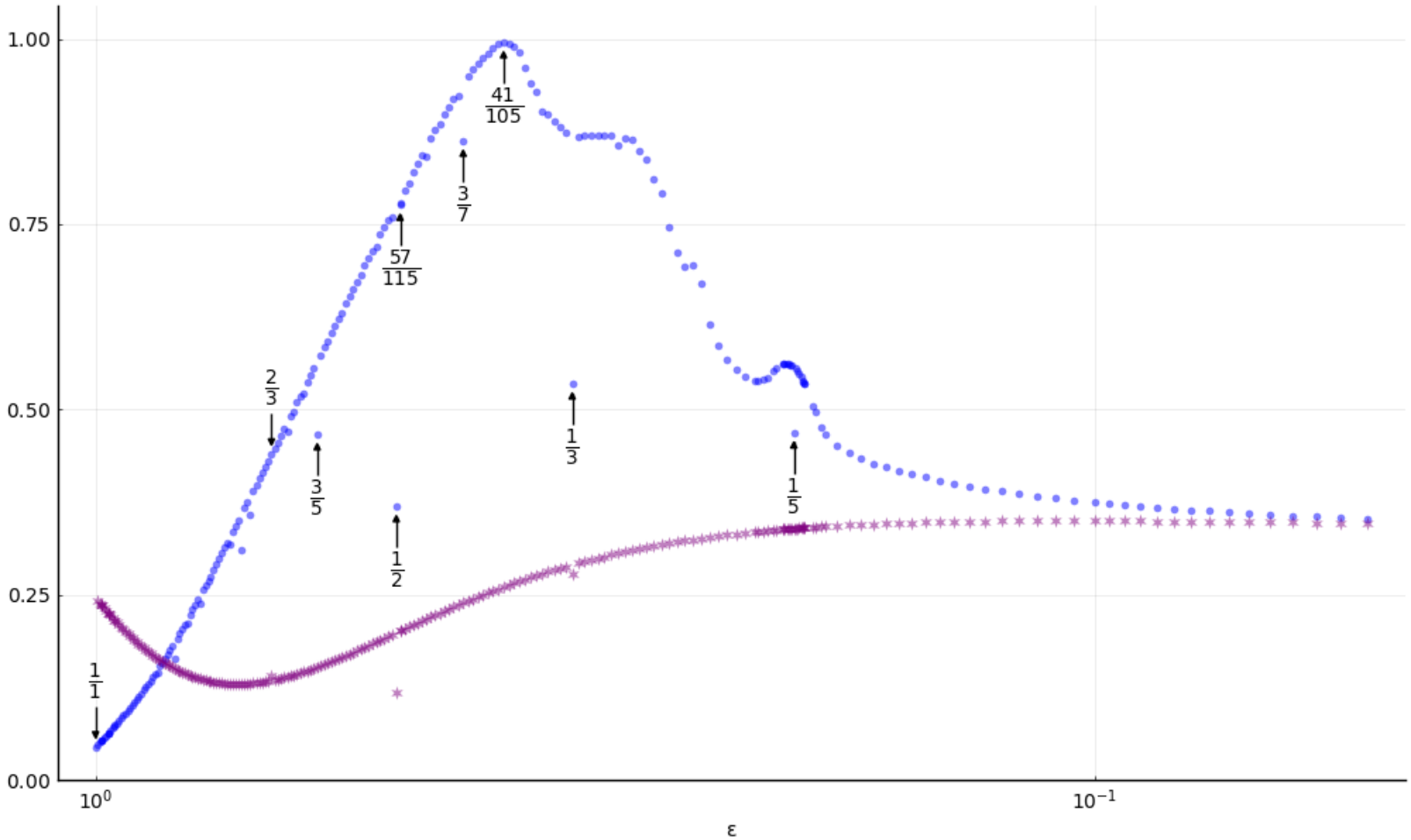


Figure: Kinetic energy of $\psi^\varepsilon$ (circles) and $\psi^{\varepsilon,(2)}_{\ell,k}$ (stars) against $\varepsilon = \frac{p}{q}$

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- Also for $\varepsilon$ not very small, order 2 is better
- Localization phenomenon for $\varepsilon$ not small
- Extension to $\operatorname{div} A\nabla$ possible